

# Work allocation challenges and solutions in a large-scale work management environment

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*There are many theoretical, technical and cultural problems which need resolving to deliver fully automated work allocation and dispatch systems. The prime aim of such systems is as a process enabler for delivery of improved quality of service with significantly reduced control and field costs.*

*This paper describes the scope of work to be allocated by the system, then highlights the main algorithm design problems and describes the overall approach taken to tackle them. This is followed by a more detailed explanation of how some of the many 'challenges' were solved using operational research algorithms and other techniques. It concludes with a summary of the results achieved to date plus the key lessons learned, together with a description of current work in hand and possible future enhancements of the algorithms.*

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## 1. Introduction

A few years ago the need for radical improvements to the allocation and dispatch of jobs to a mobile workforce was identified, and the vision of a fully automated work allocation system was proposed as part of BT's work management programme. The majority of that vision has now been realised. This paper introduces the allocation problem in section 2, discusses the overall approach taken in section 3, and provides detailed commentary on the real time algorithm and practical results in sections 4 and 5.

The key functions of the Work Manager System are:

- to automate the dispatch and closure of work to and from the field workforce,
- to determine automatically and deliver on demand the next job, making optimum use of the workforce,
- to maintain full visibility of job status and workforce,
- to provide comprehensive field-effectiveness management information statistics to support service improvements.

Work Manager provides a common solution for all BT business units [1]. Two variants exist — one for the world-wide network-facing field workforce who access the system via fixed terminals and the other for the customer-facing field workforce who access the system via hand-held dial-up terminals. The functional outline of the system is shown in Fig 1.

## 2. Scope of the allocation problem

The scope of the system includes network, residential and small business repair and provision work.

Whilst some of the work-load is known in advance and could be scheduled, the ever-tightening commitments given to customers entails an increasing proportion being same-day. The variability in task duration places great demands on effective resource management and prioritisation to achieve quality of service targets while minimising ineffective costs.

The human resources performing this work comprise the largest single component of BT's field workforce, representing approximately 20 000 field engineers and 2000 control staff. Thus, resilience and performance of the algorithms are of paramount importance. Some of the main problems solved by the scheduling and allocation algorithms can be categorised into three areas:

### Theoretical

- how to balance the many real costs (e.g. travel-time, overtime) and virtual costs (e.g. customer perception, risk of failure) associated with allocating a job, and produce the 'best' overall cost of allocating many jobs,

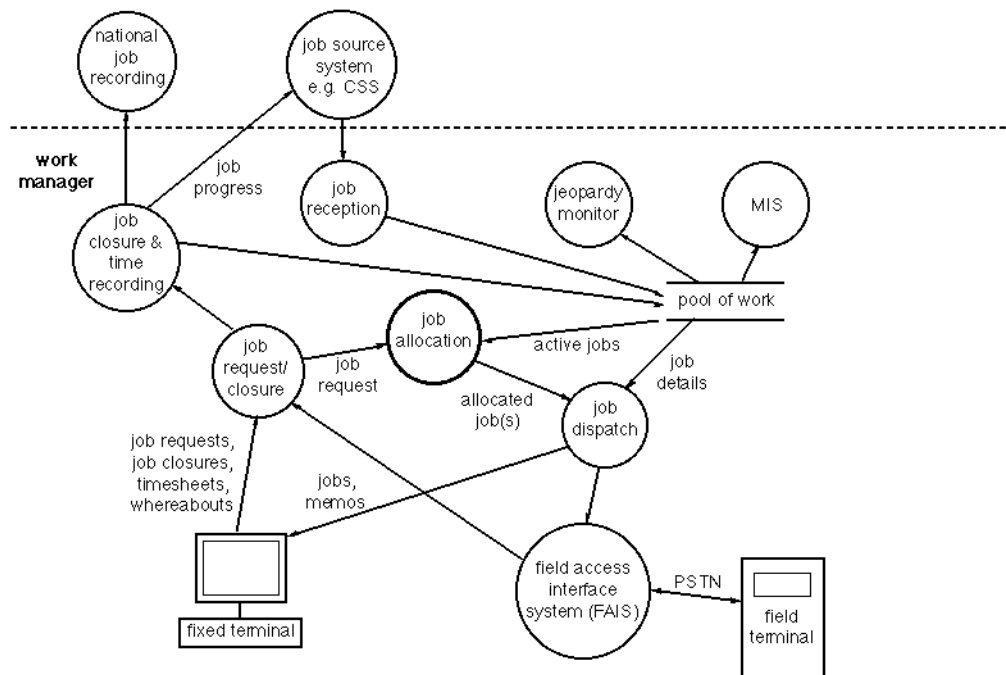


Fig 1 Work Manager overview.

- how to cope with high-priority injected work with short-term dead-lines, and, in the more general case, a dynamic/unpredictable work-load,
- how to cope with uncertain job duration and, given many appointment constraints to determine, whether to schedule and allocate a whole day's work ahead or schedule and allocate perhaps only one job ahead,
- how to balance the priorities of different types of work,
- how to estimate travel time to a sufficient degree of accuracy and cope with temporary changes such as snow, abnormal traffic conditions and roadworks,
- how to cope with a partially multiskilled workforce and the need to create work packages consisting of several short-duration jobs,
- how to provide a capability to interrupt work when appropriate to tackle high-priority jobs;
- how to provide on-line allocations within the order of 1 sec, even when engineers may request their next job significantly early, thus making the pre-allocated job either unfeasible or very sub-optimal,
- how to integrate potentially quite different algorithm techniques within one system and get them to work together where necessary;

#### Practical

- how to give visibility of current allocation proposals to give confidence to managers and administrators that the algorithms are making sound judgments given the information presented to them,
- how to make the overall algorithms flexible (e.g. parameter or rule-driven) so as to allow the allocation objectives to be changed on, or even during, the day, e.g. in storm/snow conditions to minimise travel, whilst in normal load conditions to minimise customer commitment failures (minimising travel as the second objective),
- how to determine the best rules/parameter sets to optimise performance in different operational circumstances.

### 3. Overall approach

A radical approach was taken with full automation of allocation and dispatch being the goal. However, it was still recognised that provision must be made for a small percentage of jobs which will need manual intervention/ allocation to cope with special cases.

#### Technical

- how many, and which jobs and engineers to consider, and how to constrain the solution space - given finite computing resources,
- how to quickly feed back into the algorithms important changes to jobs and engineers, without using excessive computational resources - in particular, how to cope with changes occurring whilst a new schedule or a set of pre-allocations is being calculated,

The system has the following principal components:

- a tour-build algorithm (TBA) to cope with work known well in advance,
- a real time algorithm (RTA) to cope with reactive work (typically repairs) having high uncertainty in duration,
- a travel time function to estimate travel between any two points identified by post codes (or grid references) taking into account factors affecting travel speeds between those areas,
- a next-activity function to determine if the engineer should be sent on an absence, or use a pre-built tour to get the next job, or use the RTA to allocate the next job — it also provides the capability to perform limited repair of a tour.

The way these algorithms and functions link together is shown in Fig 2.

The main differences between the two variants of Work Manager are in the network version which has no

requirement for the tour-build algorithm, but requires the following additional real time algorithm capabilities:

- a more sophisticated job selection capability to better support a partially multiskilled workforce and to ensure that even low-priority work at manned sites is visible to the algorithm (because of the much increased costs of moving engineers off sites),
- the capability to allocate several jobs at a time at the same site (because many jobs are of relatively short duration, e.g. 15 min) - so as to minimise time spent accessing the system and support some parallelism in performing jobs,
- the capability to inject higher priority work and thus prevent it from failing.

### 3.1 Tour-build algorithm overview

This algorithm deals with longer duration installation/provision jobs (e.g. 1 to 8 hours). It is needed particularly for jobs with stores requirements which must be ordered in advance. It has to be able to predict to which starting

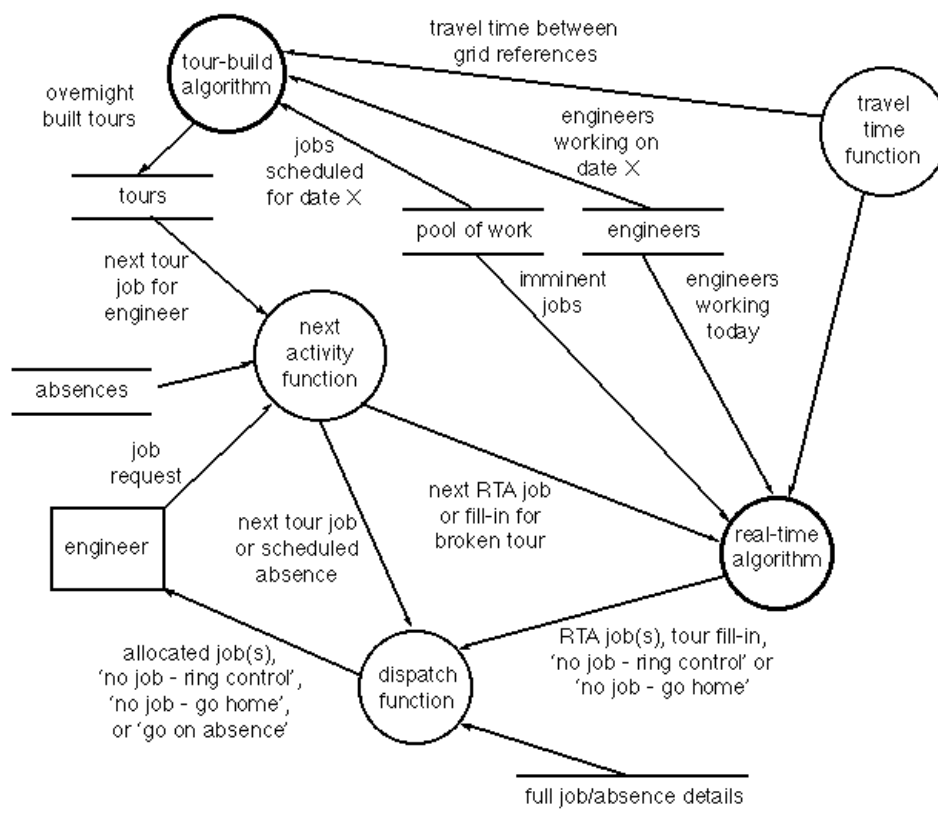


Fig 2 Algorithm relationships/interactions.

location or pick-up point the stores for a particular job must be sent (at least a day before the jobs are due to be executed). To do this a full-day scheduling algorithm was needed.

Because of its ability to deal with multiskills, the TBA is a considerable enhancement on earlier scheduling work performed within BT. It uses an expert-system approach to encapsulate knowledge of scheduling and allocation. It is written in C++ rather than using an expert-system shell because, at the time of development, it was considered that none were robust enough or capable of the necessary speed.

The approach is to subdivide the allocation problem into a set of smaller problems based on common starting points of the workforce. This is followed by assignment, to this location, of jobs which may previously have been clustered together as a viable/desirable work package. The jobs are then assigned to 'sectors' around the starting location so as to provide the basis for circular tours around the starting location. The tours are given a good initial order and the program may be repeated using randomised job orderings and relaxation of various constraints until the results no longer improve, or time runs out.

Repair and optimisation techniques are then used to fill in slack tours, improve bad tours and fit in unallocated jobs so as to minimise an overall cost function. Any remaining unallocated jobs or engineers can be put into the pool to be allocated by the real time algorithm. Figure 3 shows the structure of the tour-build algorithm.

The TBA has the benefit of being able to look ahead for the whole day but has no re-planning element and so does not handle changes on the day the tour is run. If potentially significant manual intervention is to be avoided an automated tour repair mechanism is needed - this is provided by the 'next activity' function and the real time algorithm.

The 'next-activity' function manages the running of the tours on the day. Jeopardy management processes may automatically remove jobs from the tour if they are at risk. By bringing jobs forward or using the RTA to provide a fill-in job the next-activity function will repair the tour. For this reason the RTA was first put into live operation for proving, since it provides the safety net the TBA needs to cope with the 'real world'.

### 3.2 Real time algorithm overview

For highly reactive workloads with relatively short jobs (typically 15 min to 2 hours), but great uncertainty in each

job duration, a batch-scheduling approach to determine the complete day's work for each engineer was considered inappropriate. The need for rapid response also mitigated against the use of batch schedulers, as did potentially complex schedule-repair algorithms.

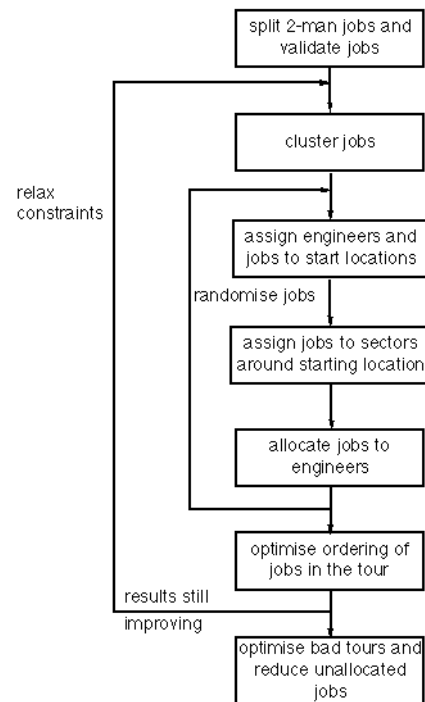


Fig 3 Tour-build algorithm structure.

In such a situation the focus of allocation concern is on which engineers are likely to be needing work in the near future and what are the top-priority jobs that need doing. The scheduling and allocation of the next job to each engineer needs to be such that the cost to the business in terms of risk (and thus the likely cost of failure) and also cost of travel and other ineffective time is minimised. As a consequence, the aim is to optimise the allocations of just the next job to be done by each engineer. This optimisation needs to be repeated on a regular basis so as to incorporate up-to-date progress information and new high-priority jobs.

The RTA is based primarily on operational research (OR) principles [1] in that the next allocation for each available engineer is found by determining the minimum overall cost in a matrix of allocation costs as shown in Fig 4.

|              | job 1 | job 2 | job 3 | job 4 | ..... |
|--------------|-------|-------|-------|-------|-------|
| technician A | 20    | 7*    | 15    | 5     |       |
| technician B | 10*   | 8     | 5     | 12    |       |
| technician C | 33    | 25    | 17    | 2*    |       |
| technician D | 51    | 33    | 11*   | 0     |       |
| .....        |       |       |       |       |       |

Fig 4 Allocation cost matrix (square matrix).

The asterisked costs represent the best allocations to produce the minimum cost. They become the pre-allocations for on-line engineers. Very small matrices can be solved by hand. However, even 10x10 matrices require considerable computation, and an efficient variation of the Hungarian algorithm [2] is needed to solve the minimisation problem.

#### 4. Detailed real time algorithm features and operation

The factors which need to be considered, when making the allocations, fall into two categories and are given below:

##### Engineer-related

- where the engineer will be next available for work,
- when the engineers are expected to be available for work,
- how much time the engineers will have available,
- what skills and access/security permissions are held,
- how quickly an engineer works relative to others,
- how far the engineer may reasonably be expected to travel;

##### Job-related

- the skills and time required to do the job with its potential variability,
- the commitment time or time window to start or complete the job,
- the static importance of the job and its dynamic importance (derived from its target completion commitment time or arrival appointment window) from which relative priority and failure costs can be derived,

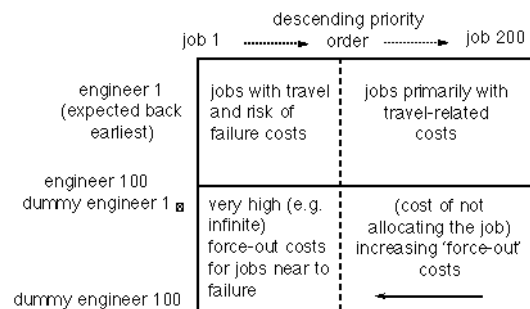
- where the job is, and whether it is within a BT building or customer's premises with constrained access.

Each of these factors can be used as either a hard constraint, producing a feasible or unfeasible allocation, or as a contributor to produce a direct cost of allocation. By valuing these costs in terms of effective travel and then adding these costs together with weighting parameters, an overall cost for a particular engineer to do a particular job can be produced. This can then be balanced against the costs of other allocation options using the matrix minimisation algorithm.

The general form of the cost equation is:

$$\begin{aligned} \text{Overall cost} = & \text{Failure Cost} + \text{Travel/Time Weight} \\ & \times (\text{Travel Time} + \text{IdleTime} + \\ & \text{Overtime} \times \text{OvertimeWeight} + \\ & \text{SkillPreference} \times \text{SkillPrefWeight} + \dots) \end{aligned}$$

The provision of user-definable weighting factors allows the business to re-focus the goals of the algorithm as business needs change. The general structure of costs in the matrix for 100 engineers in a domain with a visibility of two jobs per engineer (effectively producing an 'oblong' matrix of real engineers and jobs) is shown in Fig 5.



The matrix must be made square with 'dummy' engineers and associated job-related costs for the Hungarian algorithm to work, as described by Simpson et al [1]

#### 4.1 RTA's supporting operations

To provide the necessary on-line response times the RTA has been built into Work Manager as a proactive server which provides a set of services related to allocating a job to an on-line engineer. It also provides advance visibility of the proposed next allocations devised by its background pre-allocation process (scheduling one

job ahead for every engineer). Facilities to input important changes to engineers or jobs are also included.

To reduce the response time problem the engineers are split up into allocation 'domains' and an independent RTA runs for each domain. Jobs are then mapped into those domains on a flexible basis using a combination of job attributes to support 'functional' or 'patch' working or a mixture of the two. The relationships between the system, RTA server and its background pre-allocation algorithm are shown in Fig 6.

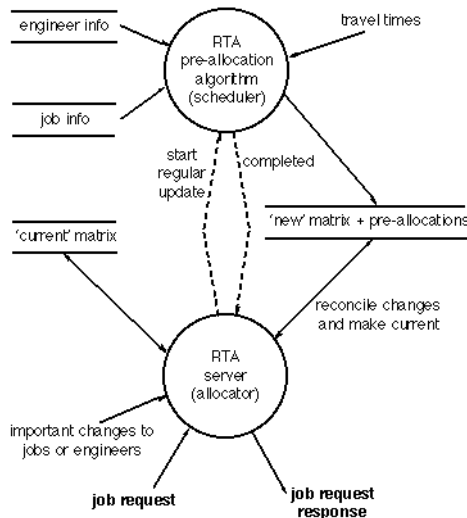


Fig 6 RTA internal structure - per RTA domain.

This structure, using a background pre-allocation process and two matrices, is determined by the need to continue giving out allocations even whilst a potentially lengthy re-calculation of new pre-allocations is taking place.

The key operations to be performed by the RTA preallocation process on a regular basis are shown in Fig 7.

Other operations are needed to deal with changes to jobs or engineers data which must be passed to the RTA immediately to prevent jobs being allocated incorrectly or non-optimally. There is a direct allocation quality/performance trade-off in this area as each change could involve re-running the Hungarian algorithm and using significant CPU power, slowing down on-line response times.

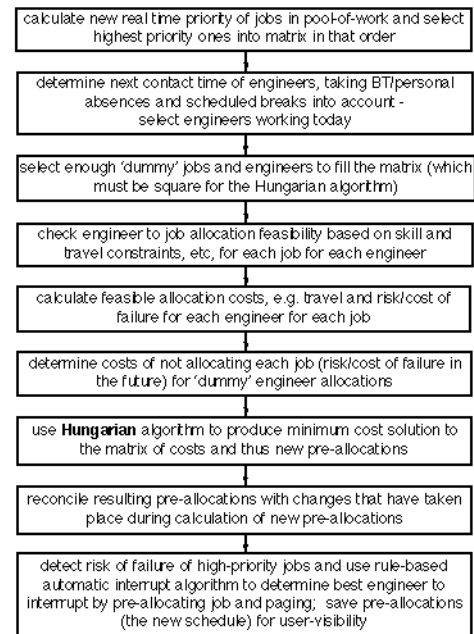


Fig 7 RTA background pre-allocation operations.

The key operations to be performed by the RTA when an engineer comes on-line to request a job are shown in Fig 8.

## 5. Results, lessons learned and future work

During the development period and early field use, the following facts were discovered which indicated future directions.

### 5.1 Real time algorithm

- The early quality of RTA allocations, when matrix sizes were restricted to looking at only one job per engineer, was operationally unacceptable. It was found that by overly focusing on key performance indicators (KPIs), e.g. avoiding appointment failures through limited priority-based job selection, travel time increased, reducing effective time and thus KPIs. The problem was overcome by increasing the visibility of jobs (seeing between two and three jobs per active engineer using larger matrix sizes) and adding costs for delaying the allocation of jobs (i.e. pre-allocating them to dummy engineers - see Simpson et al ).[1]

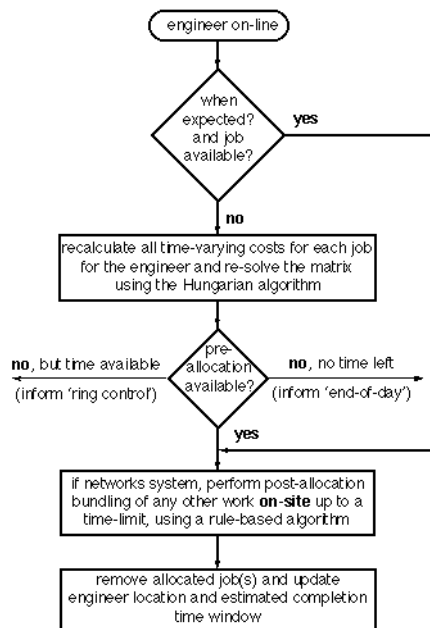


Fig 8 Engineer on-line operations.

With these changes the RTA has since proved effective at minimising travel (given fewer engineers) whilst maintaining or improving KPIs in a wide variety of workload conditions. The use of linear summation techniques for producing a balanced overall cost appears to work very well.

- In the network environment better support for multiskilling and automated job interrupts were found to be essential to meet changing business needs and could not easily be supported by the basic OR algorithm. These capabilities were added later as essentially rule-based extensions around the basic OR core. It seems increasingly the case that no one algorithm can meet all requirements and a correct mixture of techniques is essential. An area of current work is how best to deal with heavily over-constrained matrices, e.g. urgently needing to pre-allocate more jobs than there are engineers available with the right skills/mobility.
- In terms of visibility of algorithm decisions, the next pre-allocation for all engineers and their expected completion times are readily available on the system. However, it can be quite a time-consuming task to determine why engineer A was allocated job 1 in preference to engineer B or job 2. Facilities are now being tested to provide an overall feasibility/unfeasibility summary, and per engineer or job summaries of the allocation options. This will enhance

the work controllers ability to understand why particular allocations are being made when queries arise.

- There are a significant number of tuneable RTA parameters to provide a degree of future-proofing of the algorithm against business changes. Whilst only a few of these normally need to be changed to suit the work-load, experience indicated the need to be able to change parameters during the day and have the RTA respond quickly. Thus, virtually all algorithm parameters can now be re-read every 5 minutes so that the RTA responds very quickly to changes in workload. This flexibility and responsiveness has also proved invaluable in being able to make parameter changes and see the effect on pre-allocations within a few minutes. It should be noted that the costing rules themselves are fixed; however, future work could be put into a 'rule-base', thus making it much easier to try out completely new business rules.

- With many potential parameters to change, it is important to determine what set will give the best and most stable results. Extensive use of scenario testing, using real or simulated data plus test harnesses to simulate engineers coming on-line, has been used to predict the effect on KPIs and field effectiveness of changing these weighting parameters to provide optimum allocations for a given geography and resource/work-load situation.

This simulation facility also provides a powerful mechanism for predicting the impact on field effectiveness of potential business changes such as the payback from investing in greater multiskilling, or merging of workforces. Initial work has just been completed to extend this capability using a more comprehensive simulation/expert system harness called WEST (work and engineer simulation tool) [3]

- As might be expected, the algorithms are very CPU intensive and considerable effort has been required to meet tight on-line response times without resorting to the use of several super computers. Each Work Manager System runs on an 800 MIPS multiprocessor Unix box, using 1000 Mbytes of memory and 36 Gbytes of disc space. It supports up to 1400 technicians plus 140 controllers, with each of the RTA servers now supporting up to 120 engineers in an allocation domain. Up to 300 jobs can be considered for allocation at the same time.

Response times for on-line allocations are normally sub-second and even when re-calculations are required are typically within 3 sec. The regular determination of pre-allocations typically takes from 1 to 3 minutes, well within the desired recalculation period of 5 minutes. It is now being investigated whether rule-based or advanced information processing lookahead/predictive scheduling algorithms [4],[5] could meet these response times.

- Increases in allocation domain sizes are important in some network situations for reducing boundary effects, i.e. where engineers are near the edge of a domain boundary they cannot be allocated jobs just over the boundary automatically. Extensions to the algorithms to allow automated cross-domain allocations are currently being considered and would reduce the requirement for even larger domain sizes.
- Automated support for dependencies between tasks within the algorithm is limited at present and appointment windows are used as the main sequencing/coordination mechanism. Future work will concentrate on extending or augmenting the algorithms to support more complex dependencies and a greater element of forward scheduling for the important jobs. Preliminary investigations in this area and experimentation with alternative workforce-scheduling techniques have already been carried out [4],[5],[6].
- The client-server model has given the algorithms a good degree of independence from the other functions that the system must perform, and gives a clean system structure whereby other algorithms may be added to the system to work alongside existing algorithms.

## 5.2 Tour-build algorithm

- Initial testing of the TBA using live data was successful and has been used as a benchmark for comparison with other scheduling algorithms developed at BT Laboratories. Further testing to establish the best parameter settings proved that it is quite difficult to define the cost function for individual tours so that it does not conflict with the overall cost function for the complete set of tours - which must also include any jobs unable to be allocated. The introduction of multiskilled jobs also mitigates against the generation of the ideal circular (petal-like) tours.
- However, given the move within BT towards greater use of van-stocks, the pressing need to trial the TBA diminished. Other changes in policy, such as more use of multiskilled workforces (repair and installation) and

more short-term provision and installation timescales, plus the flexibility of the RTA, have led to the trialling of the RTA for provision work. As a consequence, the widespread use of the tour-build algorithm has been delayed. However, comparative tests using the scenario test harnesses such as WEST can now more readily be used to reveal the potential benefit of introducing it [3].

## 6. Conclusions

Over the last two years, the basic OR and cost-based framework used by the RTA, supported where necessary by rule-based algorithms, has proved robust as new requirements have been placed on the system.

It has not proved necessary to schedule significant numbers of jobs in advance for each engineer in order to achieve good allocations and a valuable allocation system. Live use with large populations in each allocation matrix has vindicated the choice of an OR approach for the RTA.

Future extensions to the scope of the system will increase the range of types of work and workforces supported, bringing many new challenges and algorithm enhancements, particularly in relation to greater interdependencies between jobs and stores-kitting issues.

However, the initial vision of fully automatic allocation and dispatch of work in a complex and highly dynamic work-load environment has now been realised and proved successful in extensive field trials. There are currently (October 1994) well over 2000 people being scheduled work by the system and this number is anticipated to increase to over 10 000 by the end of 1994.

## Acknowledgements

The initial real time algorithm was devised by BT's Management Science Consultancy Unit (MSCU) in London and has subsequently been greatly enhanced at BT Laboratories in conjunction with MSCU as a result of extensive feedback from users in London, Ipswich, Manchester, Bristol and the pilot sites (particularly at Leicester), without whose efforts an effective automatic allocation system could not have been developed.

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